

CMS L1T & HLT: For the LPC Trigger HATS Session, 2025

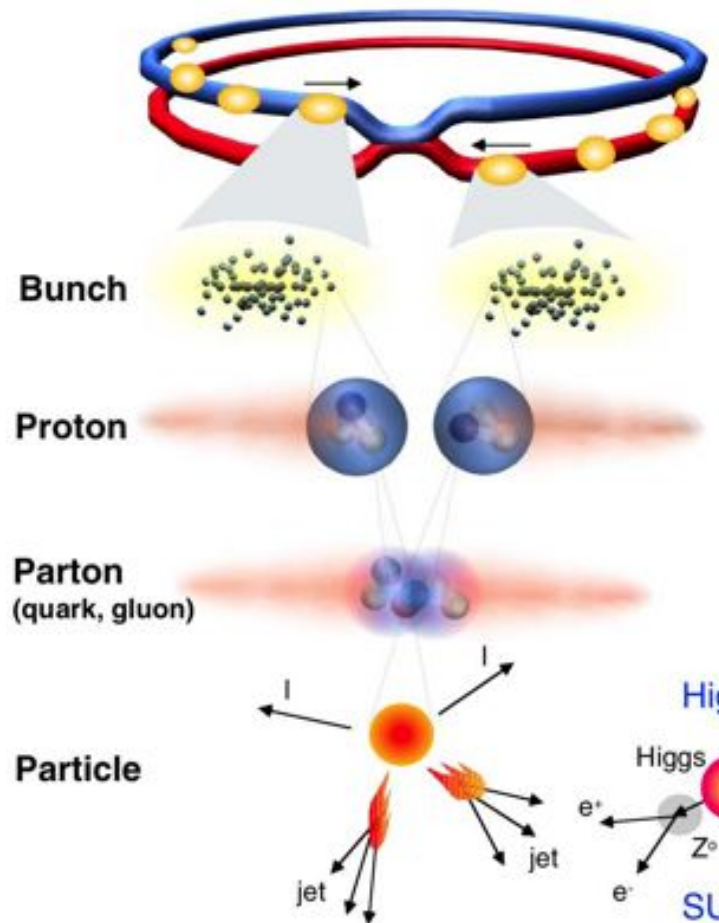
Andrew Loeliger

With material, graphics and slides taken from previous year's presentation. Credit goes to (among others): Wesley Smith, Sergio Cittolin, Tom Gorski, Ales Svetek, Maria Cepeda, Sascha Savin, Varun Sharma, Piyush Kumar, Pallabi Das, Isobel Ojalvo, Kevin Stenson, Phil Harris, Marino Misiroli

The Triggers

The Fundamental Challenge of the LHC

- Two things are needed to make any particle physics observation/discovery
 - Enough energy to make the thing you want to have happen, happen
 - Enough data showing it that, mathematically, we can't just chalk it up to random observations
- The last point cuts two ways!
 - More data -> higher statistical significance
 - More data -> more stuff to process, more events, especially background ones, to sift through and process
- What physicists (or everyone really) do when we need to find and look at a meaningful subset of observations is *build a filter*
 - Or in ML terms, a classifier
 - Try to cut away things we can label as uninteresting as soon as possible
- The design specifications of the LHC tell us what kind of filter we need to build... And it's a bit daunting



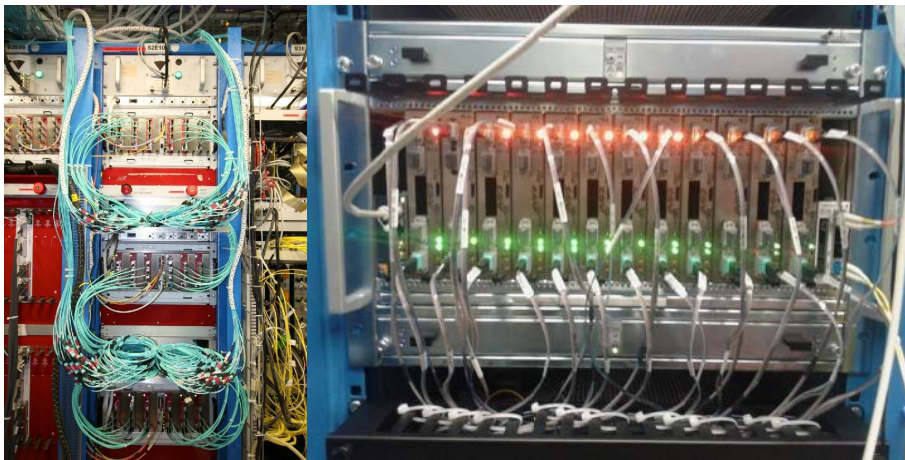
DAQ Timing and Constraints

LHC Beam Energy	7 Tev (COM energy ~13.6 TeV) (Previous record holder is out the window, at 1 TeV a beam)
LHC Orbit Frequency	~11 kHz
LHC Collision Frequency	~40 MHz (Actually closer to ~28 MHz accounting for empty bunches)
LHC Proton Bunches	4400 (Typically only ~2400-2800 “in use”, structured into “trains”)

- There is too much data coming in from the LHC to even *store* all of it
 - Full detector readout is a few MB, so multiplied by the collision frequency we get...
 - TBs a second trying to read all of it!
- We can't read all of that, but it does tell us what we need to do
 - Rate cut by 4 orders of magnitude
 - Actually done in two stages, each cutting about 2 orders of magnitude

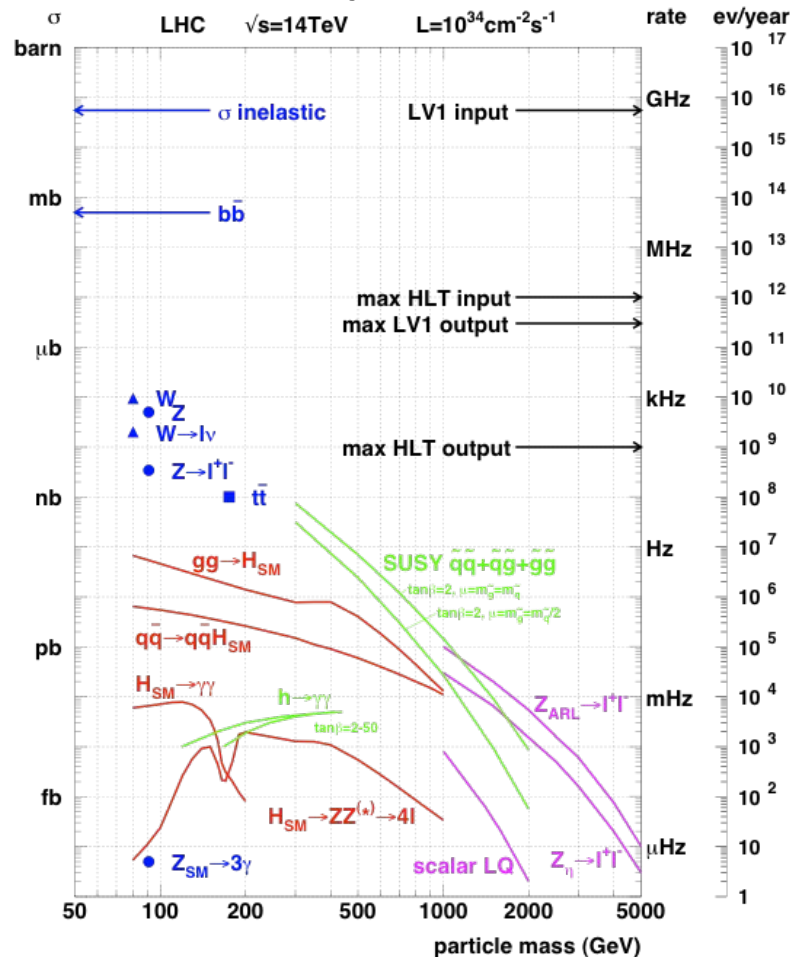
The CMS Trigger Systems: L1T & HLT

- Level One Trigger
 - “L1T”
- First pass system
- Reduced input information
- Has a “menu” of ~512 “trigger bits”, that is to say, different physics algorithms and topologies that we are looking for
 - E.g. a few simple ones that are almost always there, SingleJet180, SingleMu22, HT280
- High Level Trigger
 - “HLT”
- Second pass system on events accepted by L1T
- Full system information
- Larger menu, seeded by L1T seeds
 - Practically speaking, for your average physics analysis, this is the point of contact with the trigger, where you select the HLT trigger path you want



What do we have to be able to do with this system?

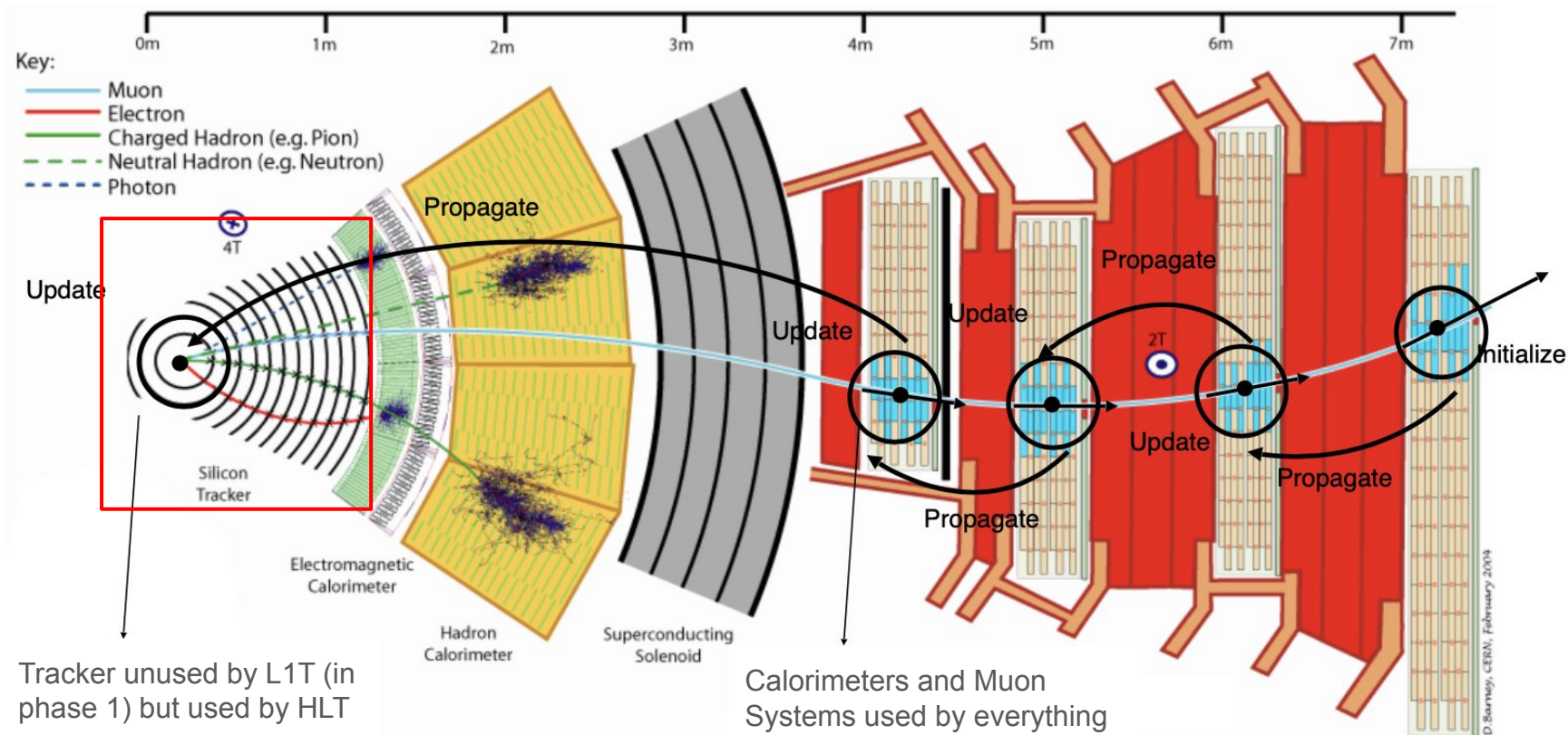
- Taken from previous slides
- Electroweak Symmetry Breaking Scale
 - Higgs (125 GeV) studies and higgs sector characterization
 - Quark, lepton Yukawa couplings to higgs
- New physics at TeV scale to stabilize higgs sector
 - Spectroscopy of new EWK produced resonances (SUSY or otherwise)
 - Find dark matter candidate
- Multi-TeV scale physics (loop effects)
 - Indirect effects on flavor physics (mixing, FCNC, etc.)
 - Bs mixing and rare B decays
 - Lepton flavor violation
 - Rare Z and higgs decays
- Planck scale physics
 - Large extra dimensions to bring it closer to experiment
 - New heavy bosons
 - Blackhole production



L1T

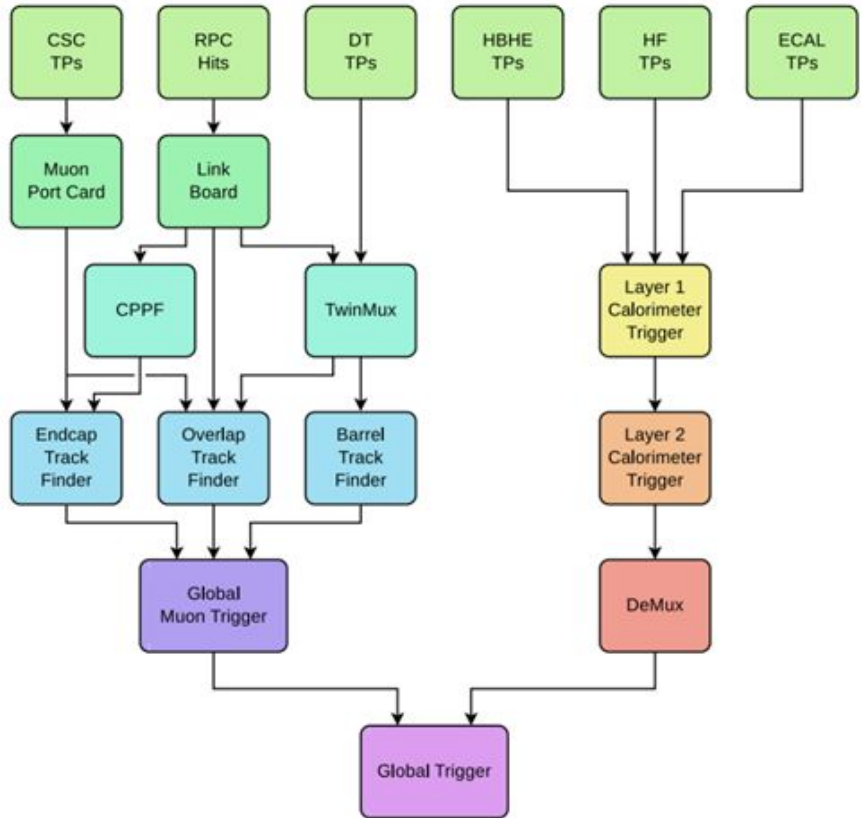
The CMS Detector(s)

If nobody has yet showed you this photo of CMS, it's very important



L1T Data Flow

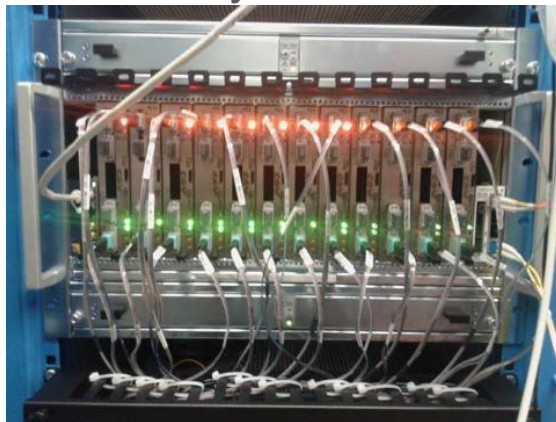
- Structure of L1T Subsystems shown on the right
- Breaks down into two “halves”
 - Calo Triggers
 - Muon Triggers
- Calo Triggers cluster objects
 - Calo Layer 1
 - Calo Layer 2
- Muon Triggers piece together tracks based on location
- Final Layer “Global Trigger” issues decision based on “menu”



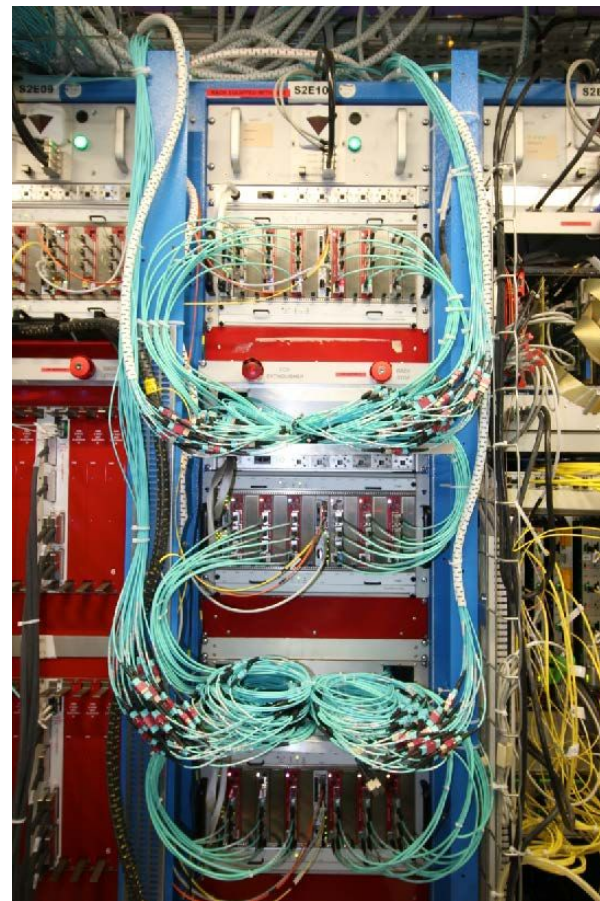
L1T Hardware and Design Specs

- L1T Is deployed on FPGAs (Field Programmable Gate Array) hardware
 - Programmable circuitry
- Targeting ~ 1 microsecond decision
- 100 kHz overall rate
- 512 distinct paths

Calo Layer 2 MP7

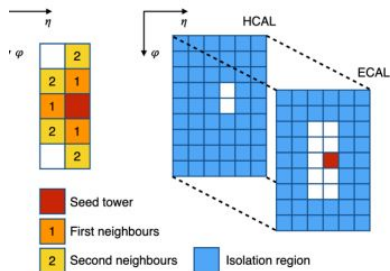


Calo Layer 1 CTP7



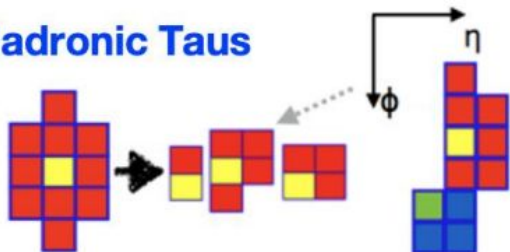
What L1T Actually Does (Part 1)

Electrons/Photons



- Clustering around seed tower with local energy maximum (> 2 GeV)
- Neighboring energy deposits clustered (>1 GeV)
- E/H and Isolation
- Energy deposit extends along the ϕ -direction

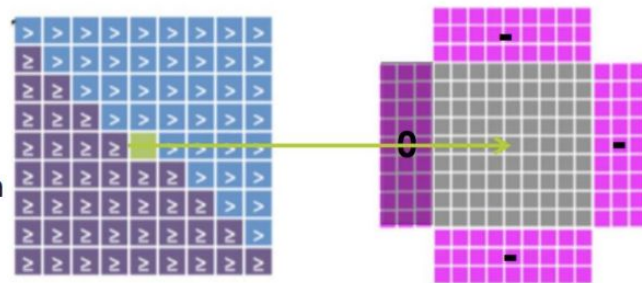
Hadronic Taus



- Clustering same as E/G
- Shape finding (Decay Modes)
- E+H and Isolation

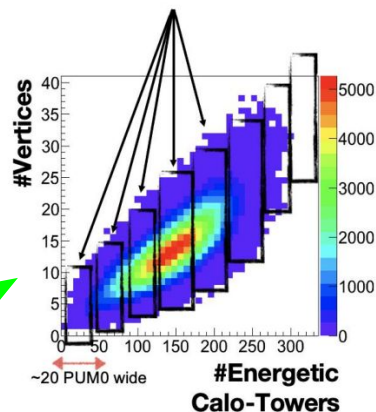
- Pileup Algorithms attempt to identify the number of p-p collisions per event and correct for the extra “noise”
- MET/Sums constructed from Summed Jets/Calo-Towers

Jets and Sums Require Regional Information



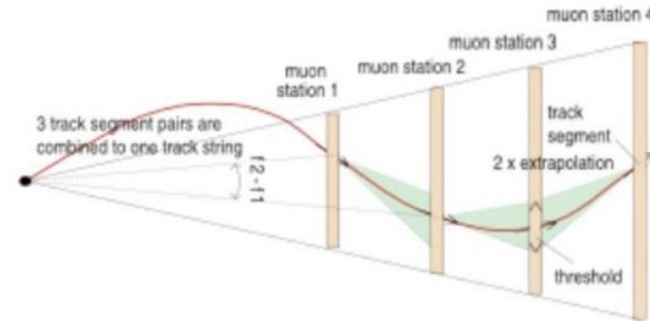
Strong Correlation

Pileup subtraction
“Chunky-Donut”



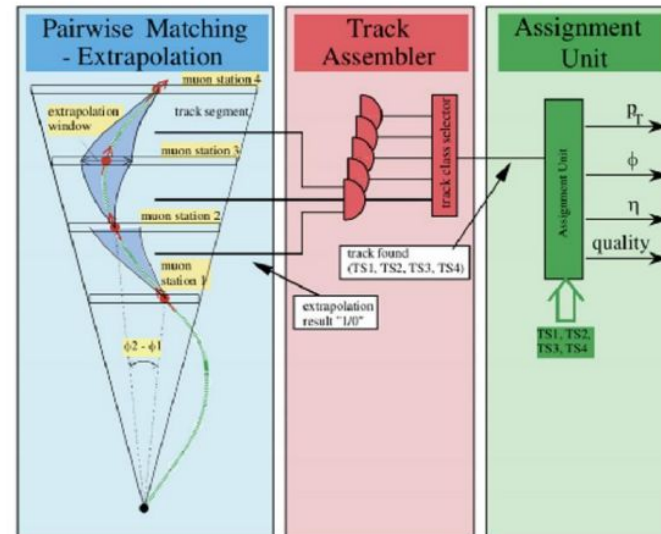
What L1T Actually Does (Part 2)

Muon Hits must be matched station by station and is dependent upon Muon Momentum

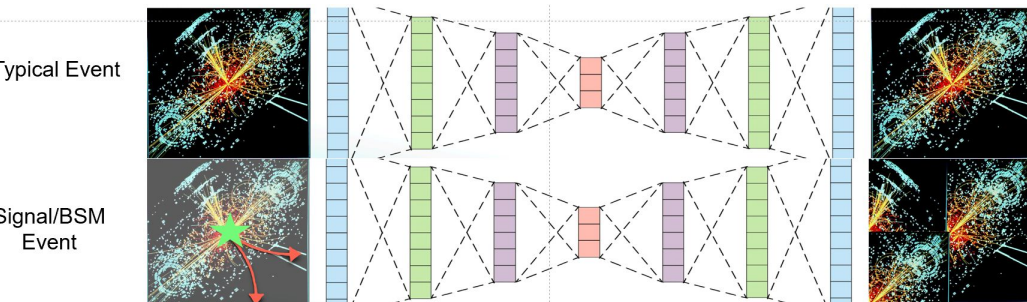


- **Input segments from Muon Chambers** forwarded to Barrel, Endcap, Overlap Muon track finders
- **Search** for track segments in adjacent modules
- **Track is assembled**
- **Kinematic assignment** based on Look Up Tables

Requires Memory to Store Patterns
Fast Logic for Matching Segments
- FPGAs are Ideal



Recent Advances in the World of L1T



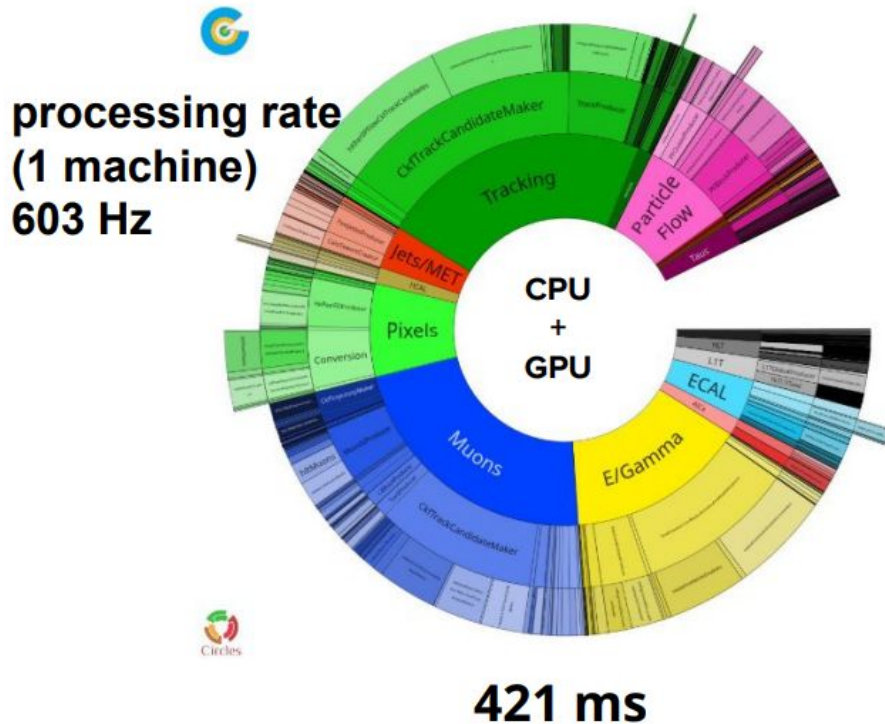
- Machine Learning and Neural Networks have come to L1T!
 - Conifer for BDTs
 - HLS4ML for neural networks
- Proof of concept: Anomaly Triggers
 - CICADA (that's me)
 - AXOL1TL
- This is how triggers are going to work going forward
 - Heap of things that had to be sorted out
 - Emulators
 - Firmware
 - Training/latency techniques
 - AXOL1TL is now *highly* focused on *pipelines*, “how do we keep models relevant, and how do we keep them coming?”

Other ideas: LLP triggers with HCAL,
Displaced muons with the muon systems

HLT

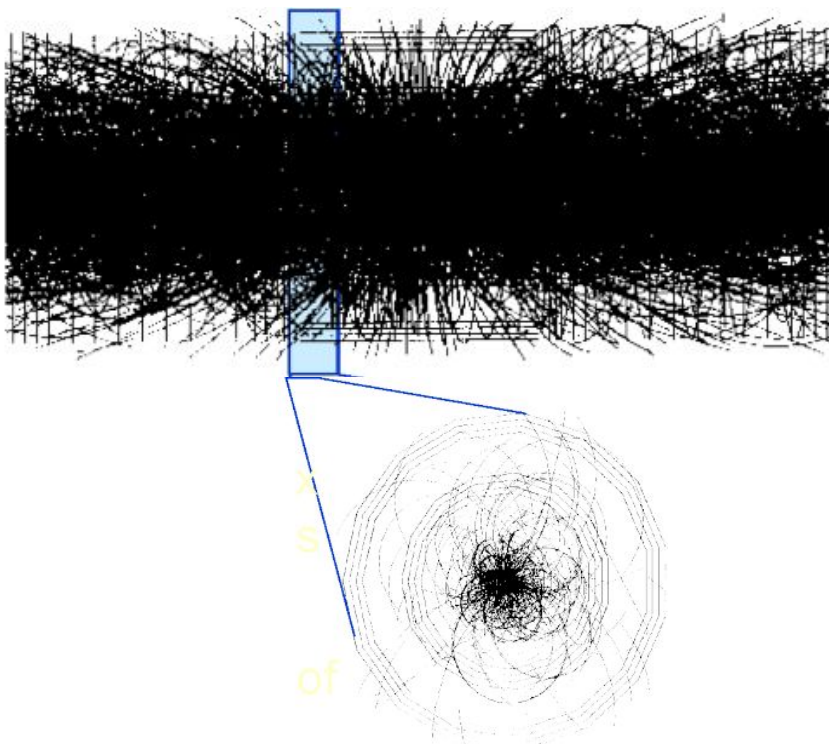
HLT Information and Operation

- HLT doesn't have distinct “subsystems” in the way L1T has
- Instead, HLT runs distributed across standard computing infrastructure
 - CPU + GPU these days
- Computing operates across paths, designed to terminate as early as possible
 - If you have any CMSSW experience, HLT is designed around it

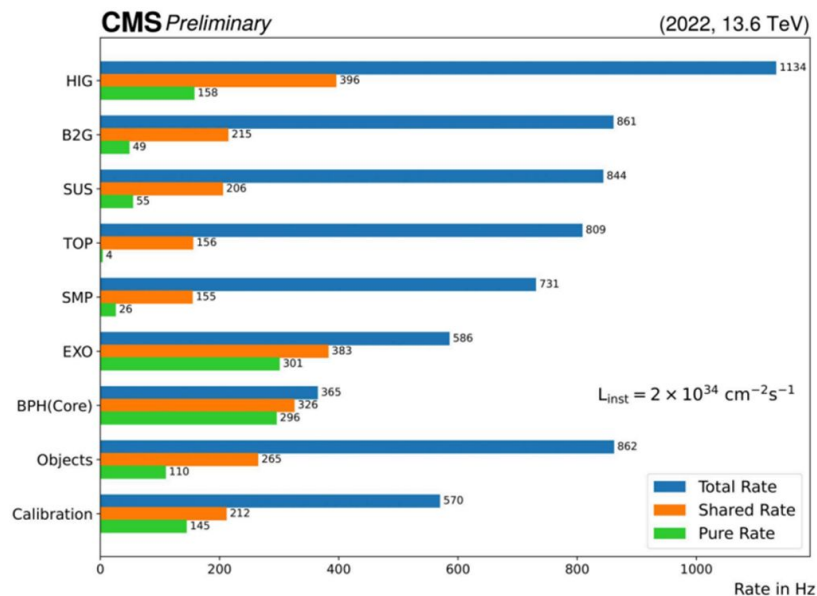


How HLT Runs

- HLT has more information than L1T to work with
 - Notably, tracker information
- Tries to run as close to offline reconstruction as can be done quickly

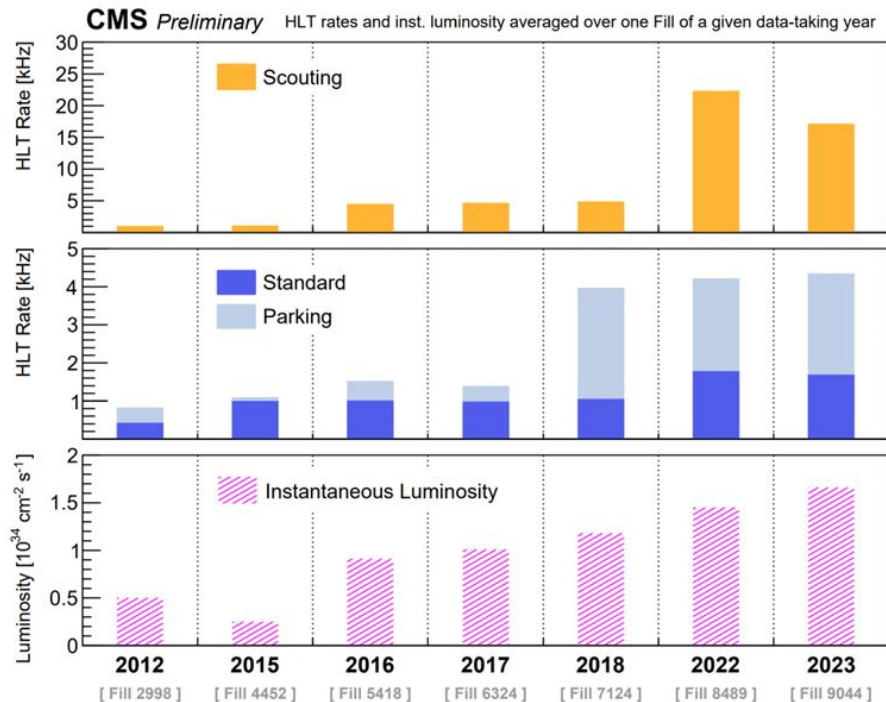


HLT rate per physics group in 2022

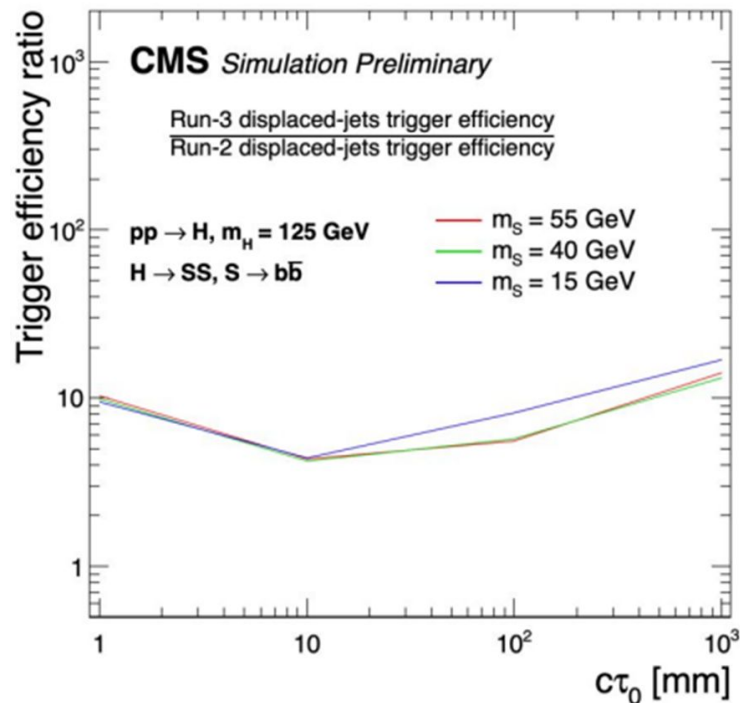


New Stuff at HLT

- Scouting
 - Trigger level analysis, i.e. pass throughs with reduced information
- Parking
 - Opportunistic reconstruction



- LLP triggers complimenting L1T
 - Plus HH and HHH
- Increasing focus on GPU reconstruction



A Quick Word About the Future: Phase 2 & The HLLHC

How CMS & Data-taking are going to change in the future

New Tracker

- Coverage up to $|\eta| < 3.5$
- Tracks available at Level 1 Trigger
- Radiation tolerant - high granularity
- less material

Muons

- Replace DT FE Electronics
- Complete RPC coverage in forward region (new GEM/RPC technology)
- CSC replace FE electronics for inner rings

Barrel ECAL

- Replace FE Electronics
- Crystal-level information at L1

Barrel HCAL

- Replace HPD by SiPM

New Endcap Calorimeters

- High granularity (HGCAL)
- Segmented depths

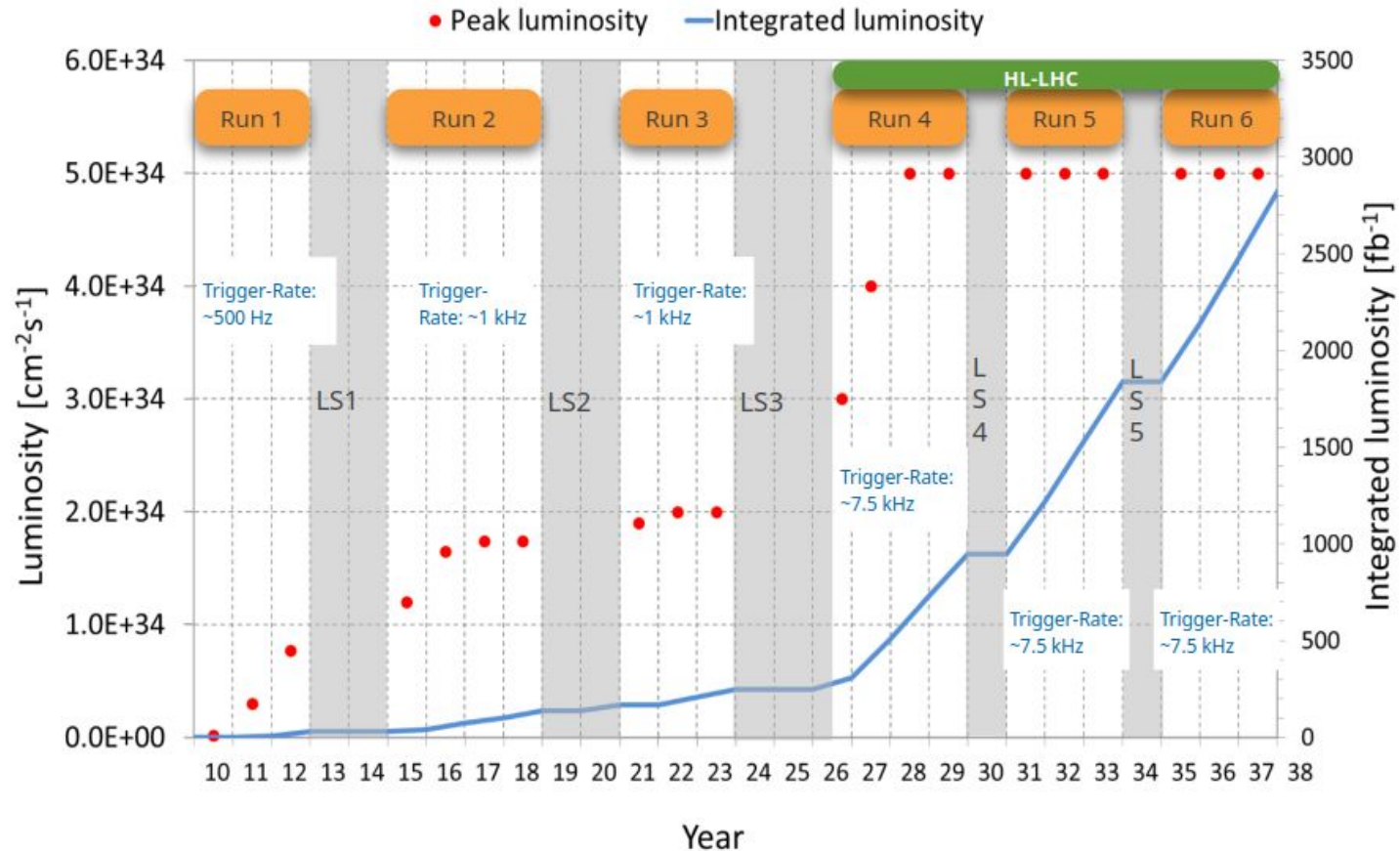
Trigger/DAQ

- Tracking and ECAL Crystal information available
- Rate up to 750 kHz
- L1 Latency up to 12.5 μ s
- HLT output rate up to 7.5 kHz
- New DAQ hardware

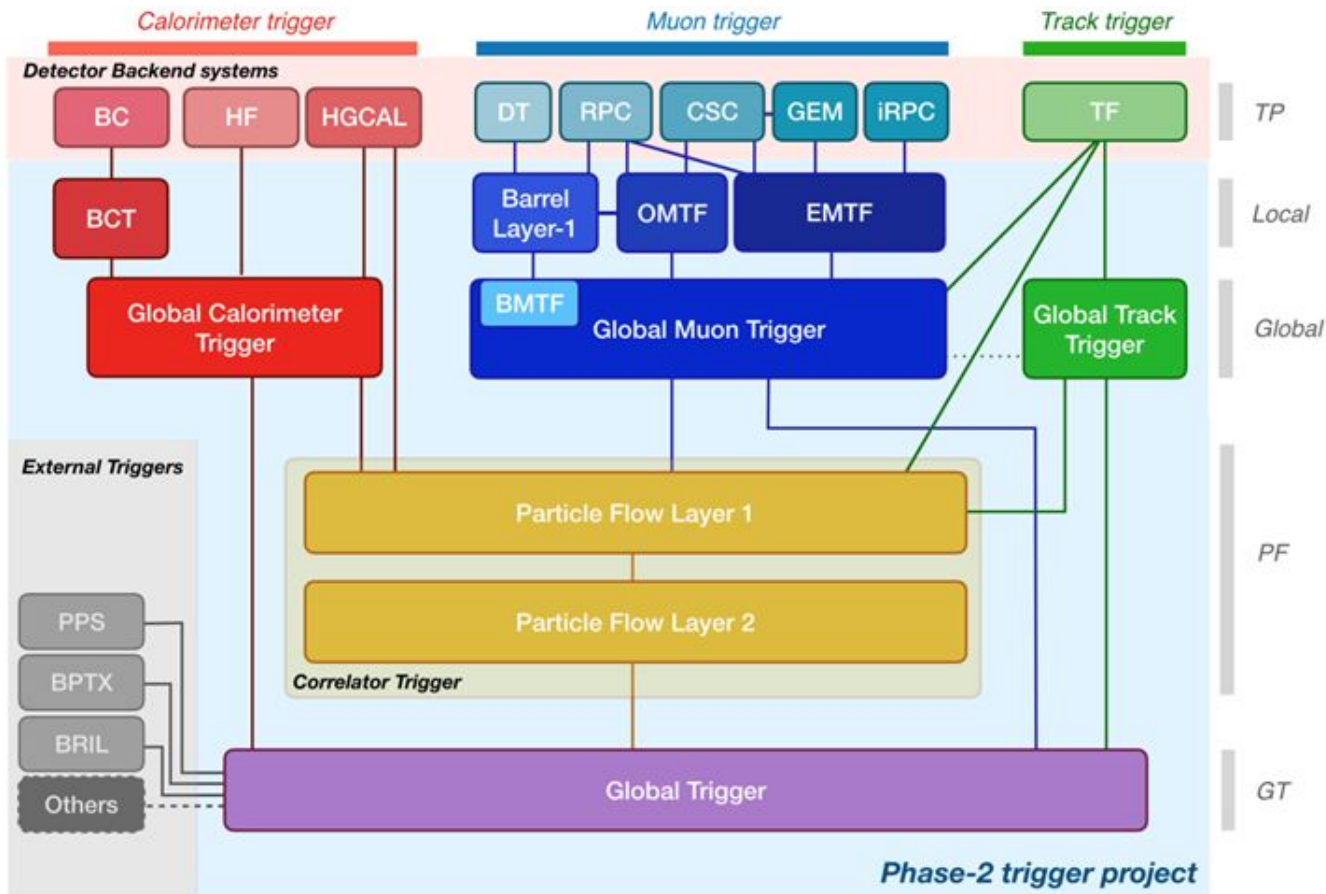
New Timing Layer

- Thin detector outside of tracker
- Timing Resolution ~ 30 ps

Whole new challenges in triggering and data-taking



How the triggers are adapting



Practical Elements of the Trigger for Everyone

Trigger Emulation

- The triggers are not just online software and hardware, they are offline software as well
 - In particular L1T has an entire emulator
 - [GitHub](#)
- cmsDriver has dedicated L1T and HLT steps
 - [\(old\) recipes can be found here](#)
- Vital for MC and studying the trigger
- Should also be clear on the difference between emulator results and “unpacked results”
 - Both show up in different places as ways of studying the trigger
 - Emulator is more common, it is things our emulator puts out and says L1T should have done
 - Unpacked can show up in formats and Ntuples derived from RAW, and it is after the fact reconstructions of the information L1T sent between itself (this is saved as a matter of policy for L1T debugging and studying)

Convenient Trigger Studying

- Speaking as someone who has had to do it, studying the trigger, it's objects, and results has been one of those things that has discouraged people in the past
- But fret no longer! It has gotten markedly easier over the last few years, and I am here to help you get the information you need!
- L1T
 - The fundamental difficulty with studying L1T is that we just don't keep much L1T information past a certain point
 - Global trigger objects and results... which is what most people need
 - In case you need more, L1T maintains an old common Ntuple format, [L1TNtuples](#)
 - Trick is, it runs on RAW
 - [But the new method is L1Nano](#)
 - Addition to the popular NanoAOD format, easy to work with flat Ntuples
 - Will get you object and trigger bit info, along with stuff earlier in the chain
- HLT
 - HLT is considerably easier. Bit/path information is by necessity in almost every data format
 - Objects should be close enough to offline objects that you can just study those I think
 - Not an expert on this admittedly.

“I want a new trigger for my analysis”

- Again, speaking as someone who works in this world, the trigger can be intimidating and difficult enough to reckon with that analysts simply don't want to try and figure out changing the trigger to support new ideas and new physics searches
- But they should!
 - If your signal doesn't get caught at the trigger, it doesn't get caught period!
 - In my opinion, the trigger groups, especially L1T doesn't get enough interaction from analysts who want to try new and interesting things
- So how do you get a new trigger in?
 - Depends on where you want it.
 - Unfortunately, I can't speak too much to the HLT side of things. The general set of steps is that you can modify an existing menu to have the seed you want (based on HLT objects), and you will need to demonstrate that it doesn't exceed HLT's timing budget
 - L1T side, [we have a dedicated menu editor](#) that you can edit an existing menu to have the seed you want. Then similar to HLT you will need to re-emulate performance on data, and check the rate and efficiency.
 - If you need something that is not the standard objects, that is a more involved project
- The triggers do hold their own dedicated tutorials for this sort of thing (or they have in the past)

Trigger Efficiency and Scale Factors.

- And this brings us towards things that will soon be exercises
- The most common touch point for analysts with the trigger is needing to do efficiency studies and derive scale factors
- Common use case(s):
 - My trigger is 96% efficiency on the objects of interest in data... but how about in MC? You will need to derive scale factors
 - Tag and Probe Method
 - Where does the trigger “turn-on”? I.e. where do you actually cut to get a signal enriched region?
 - How about on signals or overall?
 - Don't have convenience of the other methods? Find triggers that are orthogonal
 - Orthogonal trigger method
- I'll point to the trigger slides [in this old set of slides](#)

Some final thoughts

Final Thoughts

- The trigger system really is the whole detector in mini
 - It touches nearly everything the detector does:
 - Hardware
 - Software
 - Reconstruction Techniques
- We have a lot of exciting things going on these days at the trigger
 - ML is the new big thing at L1T
 - HLT is expanding into GPUs and new ways of considering how we take data (different reconstruction times and data bandwidths)
 - New physics goals and benchmarks!
 - And Phase 2 is just around the corner!
- More than ever, with the complexity we are introducing earlier and earlier into the system, we need engagement from people who understand the trigger system, and have clear physics cases.
- Let's do some exercises using the trigger!